2024 International Conference on Smart Applications, Communications and Networking (SmartNets) | 979-8-3503-8532-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/SMARTNETS61466.2024.10577646

Traffic-Aware Network Slicing for 5G Networks in Cloud Fog-RAN over WDM Architecture

Muhammad Ahsan
SnT Center
University of Luxembourg
& NUST, Pakistan
muhammad.ahsan@uni.lu

Ashfaq Ahmed
Comp. & Commun. Eng. Dept.
Khalifa University
Abu Dhabi, UAE
ashfaq.ahmed@ku.ac.ae

Sabit Ekin
Eng. Tech. & Ind. Dist. Dept.
Texas A&M University
Texas, USA
sabitekin@tamu.edu

John O'Hara
Elect. & Comp. Eng. Dept.
Oklahoma state university
Stillwater, USA
oharaj@okstate.edu

Arsalan Ahmad
Elect. & Comp. Eng. Dept.
Iowa state university, USA
& NUST, Pakistan
aahmad@iastate.edu

***Abstract*—Fifth generation (5G) mobile networks will utilize network slicing to support a diverse range of services, each with distinct specifications for data rate, latency, and reliability. These services are broadly classified into 3 main categories: ultra-reliable low-latency communication (URLLC), enhanced mobile broadband (eMBB), and massive machine type communication (mMTC). Network slicing efficiently provisions these services by creating multiple virtual networks over the same physical infrastructure using network function virtualization (NFV). In this paper, we propose a traffic-aware network slicing scheme by levaraging a 3-layer cloud fog radio access network (RAN) (CF-RAN) over wavelength division multiplexing (WDM) architecture. This network architecture consists of remote radio heads (RRHs) at the first layer, fog nodes at the second layer , and baseband unit (BBU) hotels at the third layer. The fronthaul for incoming URLLC requests are served at the fog nodes because of their stringent latency requirements, while the fronthaul for eMBB and mMTC requests is handled at BBU hotels. The proposed architecture dynamically adapts to the random traffic generated by each RRH. Furthermore, we propose a heuristic algorithm to investigate how changes in the service request ratio effects the overall performance of the proposed architecture. The algorithm minimizes resource utilization while maintaining maximum network connectivity. The results demonstrate the impact of incoming slice request variation on the number of active nodes, number of served requests, and aggregated fronthaul delay.**

***Index Terms*—5G, Network Slicing, CF-RAN over WDM, URLLC, eMBB, mMTC.**

I. Introduction

THE goal of 5G and beyond networks is to provide unprecedented performance for a variety of services, including autonomous driving, remote surgery, 8K video streaming, etc. These services have been categorized as URLLC, eMBB, and mMTC [1], and they place heterogeneous data-rate, latency, and reliability demands on the networks [2]. URLLC includes applications with stringent end-to-end (E2E) latency ($\leq$ 1ms) and reliability ($\approx 99.99\%$) requirements, including intelligent transportation, industrial automation, and remote surgery. The eMBB services are more bandwidth-intensive, such as 8K video streaming, and augmented/virtual reality services. The mMTC services permit massive connectivity of internet of things (IoT) devices at relatively low data-rates. Network slicing in 5G networks incorporates all these services with divergent requirements by establishing multiple virtual networks over the same physical network using NFV [3], [4]. Network slicing is a promising technique for handling rapidly expanding applications with specific requirements.

In recent years, many researchers have worked on implementing network slicing in cloud RAN (C-RAN) architecture [5]. In C-RAN the BBU hotel are separated from the RRH and are consolidated at a common location where they can serve multiple RRHs [6]. Even though the centralization of BBUs in C-RAN promises efficient resource allocation and cost savings [7], [8], it cannot effectively support services with diverse requirements due to scalability issues. Consequently, the most recent research has focused on network slicing in 3-layer radio unit (RU)-distributed unit (DU)-central unit (CU) architecture, in which baseband processing functions are further divided between DU and CU [9]. Since the DU and CU functions are handled by central offices (COs), adapting to variations in incoming traffic becomes challenging. This limits the benefits of this architecture over the traditional 2-layer C-RAN architecture [10].

To address the challenge due to random behavior of incoming slice requests, we propose a 3-layer CF-RAN over WDM architecture consisting of RRH at layer 1, fog nodes at layer 2, and BBU hotels at layer 3. Fog nodes are positioned closer to the RRHs at cell sites (CSs) and are dedicated to serving delay-sensitive URLLC requests. Unlike RU-DU-CU architecture, entire URLLC requests are processed at the fog nodes to meet their strict latency and reliability requirements. Moreover, the eMBB and mMTC requests are processed at the BBU hotels. The traffic between RRHs and fog node/BBU hotels constitutes the fronthaul traffic carried by enhanced common public radio interface (CPRI) (eCPRI) using split option 7.2 as opposed to split option 8 in C-RAN that has huge data

979-8-3503-8532-8/24/$31.00 © 2024 IEEE

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.

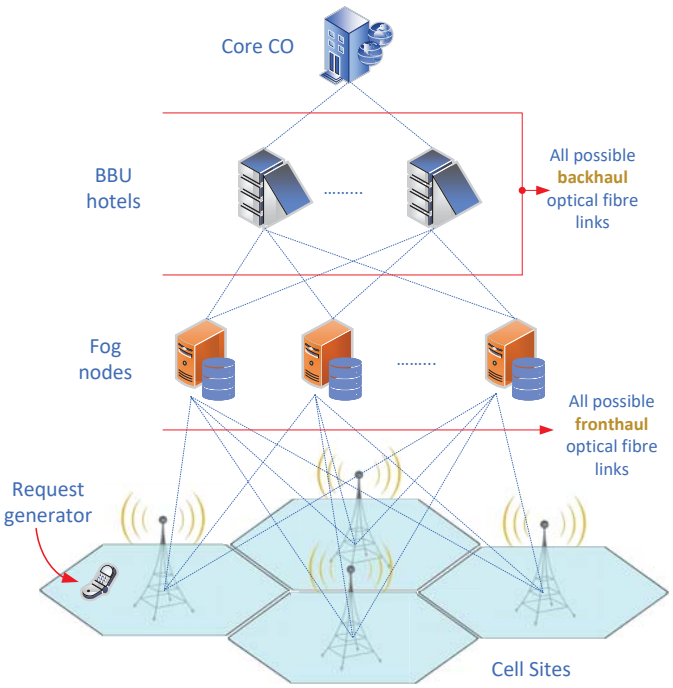


Fig. 1: CF-RAN over WDM architecture.

rate requirement. The backhaul traffic is transported between the fog node/BBU hotels and the core CO. Fog nodes have limited processing capability, which simplifies their integration to the network. This article evaluates the performance of the proposed CF-RAN over WDM architecture with respect to random traffic behavior for realistic network scenarios.

A. Literature Review

In [11], the authors discussed the motivation and challenges associated with 5G network slicing implementation. In [12], an algorithm is proposed to support a large number of IoT devices and rapidly increasing user demands in C-RAN. Domenico *et al.* [13] investigated the virtual network functions (VNF) placement and allocation of computational resources while deploying hybrid cloud C-RAN architecture. Furthermore, [14] formulates a mathematical model to minimize E2E energy consumption in virtualized C-RAN. The authors in [15] propose an optical-wireless integrated fronthaul concept to support eMBB and mMTC services while minimizing overall cost. The authors in [16] presented a heuristic algorithm to enable isolation among diverse network slices in the 3-layer RU-DU-CU architecture. Song *et al.* [17] provided an hierarchical edge cloud based architecture and analyzed the E2E delay and throughput for URLLC, eMBB and mMTC services. In [18], an integer linear programming based model is formulated that minimizes cost through optimal placement of RUs in optical RAN.

B. Motivation & Contributions

According to the literature review, the majority of research work focuses on efficient resource allocation and cost savings while deploying C-RAN and RU-DU-CU architectures. To the best of our knowledge, no published work has explored the impact of random traffic variations for different network slicing while deploying CF-RAN over WDM architecture. The main contributions of this article are as follows:

- We investigate the impact of random behavior of incoming URLLC, eMBB and mMTC traffic on the overall performance of the network.
- We present a heuristic algorithm for evaluating the impact of traffic variation on CF-RAN over WDM [19]. The objective of the algorithm is to serves maximum number of incoming requests by activating minimum number of fog nodes/BBU hotels while considering the constraints for fronthaul latency and link capacity. Furthermore, the algorithm performs grooming routing and wavelength assignment (GRWA) for all three types of incoming requests.

C. Paper Organization

The rest of the paper is organized as follows. In Section II we will explain the proposed network architecture along with the types of traffic and the types of functional split option defined by third generation partner project (3GPP). The proposed heuristic algorithm is presented in Section III. The results are discussed in Section IV, and finally the paper is concluded in Section V.

II. Network Architecture

Our 3-layer CF-RAN over WDM architecture is shown in Fig. 1. The nodes are arranged in 3 layers with CSs at layer 1, fog nodes at layer 2, BBU hotels at layer 3. On top of it is the core CO. All nodes are interconnected through optical fiber links. Each fiber comprises multiple wavelengths, and each wavelength is assigned to a particular service type to ensure isolation. Moreover, In our proposed architecture, the computing tasks for URLLC requests are performed locally at the fog node, while for the eMBB and mMTC requests they are performed at the BBU hotel.

A. Traffic Types

In this work, the URLLC, eMBB and mMTC are referred to as type 1, 2 and 3 requests, respectively. It is assumed that only the same traffic type can be groomed into shared lightpaths (optical channels spanning one or more physical links) to ensure isolation between the slices [20]. For type 1 requests, the fronthaul traffic is carried between the RRHs and fog nodes, while backhaul is transported between the fog nodes and the core CO. The backhaul request of URLLC traverses the BBU hotel node without being processed. The fronthaul traffic for type 2 and type 3 requests is processed at the BBU hotels as it traverses fog nodes without being processed, as fog nodes are equipped with reconfigurable optical add-drop multiplexer (ROADM) [21]. Moreover, the backhaul for type 2 and type 3 requests is carried between BBU hotels and core CO same as in traditional 2-layer architecture.

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.

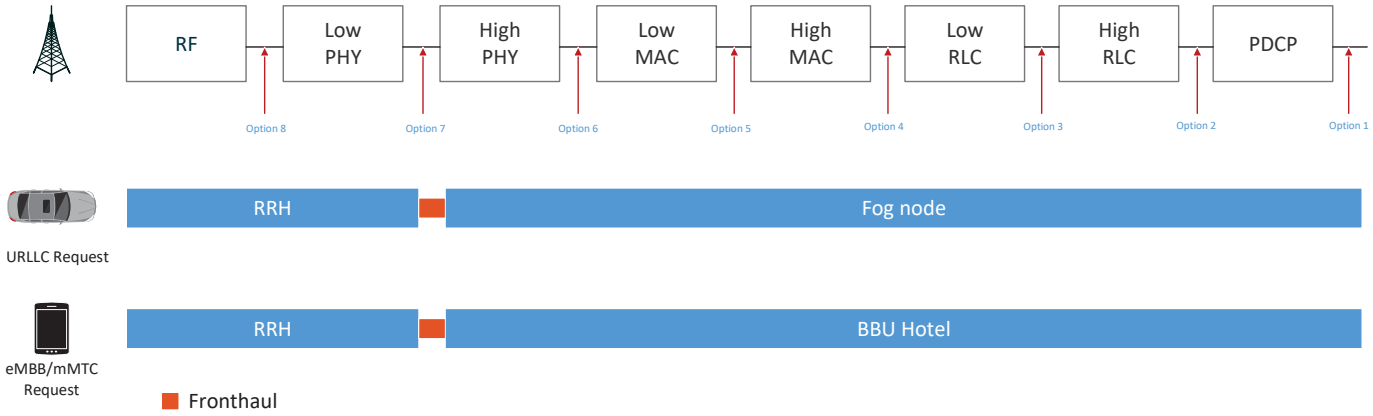


Fig. 2: Adopted functional split point between RRH and fog node/BBU hotel for URLLC, eMBB and mMTC requests.

B. Functional Splits

The functional split is the splitting of functionalities between the RRH and the BBU hotel. The employed split option at the RRH decides which functions are performed at the RRHs and which are centralized. 3GPP defines 8 split options, each of which has distinct data-rate and latency requirements [22]. Traditionally, C-RAN used split option 8, which offers the highest degree of function centralization when compared to all other options, as can be seen in Fig. 2. However, it has unrealistic fronthaul data-rate requirements, making it an unfeasible option [7]. Split option 7 is currently the most popular for the fronthaul transport in 3-layer RU-DU-CU architecture [23]. It provides significant centralization gain and has a less stringent fronthaul data-rate requirement than split option 8. In this work, we have therefore considered the split option 7 for the transport of fronthaul traffic between RRHs and the fog nodes/BBU hotels. In addition to fronthaul traffic, the RU-DU-CU architecture also considers midhaul traffic routed through split option 2 between DU and CU [16]. Midhaul traffic is not considered in the proposed CF-RAN over WDM architecture, as total processing of all slice requests occurs at the RRH and fog nodes/BBU hotels.

III. Traffic Aware Joint BBU Hotel and Fog Node Activation Algorithm (TAB-FAA)

We present a heuristic algorithm named TAB-FAA, as illustrated in Algorithm 1. The notations used in the proposed algorithm and its corresponding discussion are summarized in Table I. The algorithm determines the locations of minimum number of active BBU hotels and fog nodes based on the varying proportion of incoming slice requests. Moreover, it executes GRWA on all requests while isolating each slice and maximizing network connectivity. In the algorithm, $\mathbb{N}$, $\mathbb{R}$ and $\mathbb{T}$ denote the set of nodes, requests and request types, respectively. The core CO is denoted by o. The maximum allowable fronthaul delay for type t is denoted as D_t. The request types are given as $t \in \mathbb{T} = \{1, 2, 3\}$, where $t = 1$ represents URLLC, $t = 2$ represents eMBB, and $t = 3$ represents mMTC requests types. The required fronthaul and backhaul capacities for type t request are denoted as $C_F^{(t)}$ and $C_B^{(t)}$, respectively. Finally, the path between two nodes is represented as P_{ab}.

Incoming requests in set $\mathbb{R}$ are partitioned in predefined proportions for all slice types. A certain request r is identified by its generating node s and its type t (**Line 2**). If r is a type 1 request and s is associated with an active CO, the distance between s and the active COs is calculated. The path delay Q is then calculated using the calculated path length (**Line 6**), where total delay includes propagation delay, transmission delay, interface delay due to the used split option at RRH, and electronic switch delay. The path is used to route r if $Q \leq D_t$ and the path's link capacity is sufficient for routing r (**Lines 8-9**). If r cannot be served at an active CO, residual capacity is calculated in the link between s and the fog node associated with s, as well as the link between fog node and core CO o. If sufficient capacity is available, the fog node to which s is connected is activated if it is not already active, and r is carried along the computed path (**Lines 14-15**). If r cannot be satisfied at the fog node, then request is blocked (**Line 17**).

If incoming request r is a request of type 2 or type 3, and s is not associated with an active BBU hotel, then all reachable COs from s are examined. After that, the shortest path between the candidate CO and s is determined using the Dijkstra algorithm, and the delay of this path is calculated (**Line 22**). The required fronthaul latency is calculated by path length. If Q is less than or equal to the maximum allowable fronthaul latency and the remaining link capacity is sufficient to carry fronthaul and backhaul traffic for r, then s is associated with the CO. If candidate CO does not have an active BBU hotel, it is activated and r is routed via the computed path (**Lines 24-25**). Moreover, r is blocked if no CO can host BBU for s (**Line 27**). If source s is already associated with an active BBU hotel upon the arrival of r, its location is determined (**Line 29**). If the residual capacity along the determined path is sufficient to transport $C_F^{(t)}$ and $C_B^{(t)}$, then r is transported via the computed path (**Line 31**); otherwise, r is blocked (**Lines 33**). The process is repeated for all incoming requests.

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Traffic Aware Joint BBU Hotel and Fog Node Activation Algorithm (TAB-FAA)

Input: Network topology, $\mathbb{N}$, $\mathbb{R}$, $\mathbb{T}$
Initialization: $flag \leftarrow 0$
Output: No. of active BBU hotels, No. of active fog nodes

foreach *request* $r \in \mathbb{R}$ **do**
  $s \leftarrow$ source generating r, $t \in \mathbb{T} \leftarrow$ type of r
  if $t == 1$ **then**
    URLLC request
    foreach *reachable CO from* s *with BBU hotels* **do**
      $Q \leftarrow$ path delay from s to CO
      if $Q \leq D_t$ *& capacity available from* s *to core CO* **then**
        route r on path P_{so}
        $flag \leftarrow 1 \implies$ **break**
    if $flag == 0$ **then**
      foreach *fog node* **do**
        if s *is connected to current fog node* **then**
          if *fronthaul and backhaul capacities available* **then**
            activate fog node for s
            route r on path $P_{so} \implies$ **break**
          else
            return request-blocked
  else
    eMBB / mMTC request
    if *no BBU activated for* s **then**
      foreach *reachable CO from* s **do**
        $Q \leftarrow$ delay of shortest path from s to CO
        if $Q \leq D_t$ *& capacity available from* s *to core CO* **then**
          activate BBU in CO and link to s
          route r on path $P_{so} \implies$ **break**
        else
          return request-block
    else
      find BBU linked to s
      if *fronthaul and backhaul capacity available* **then**
        route r on path P_{so}
      else
        return request-block

TABLE I: List of notations used in Algo. 1.

Notation	Explanation
$\mathbb{N}$	set of nodes
$\mathbb{R}$	set of requuests
$\mathbb{T}$	set of requests' types
r	index of a request
t	request type
o	core CO
s	source node
D_t	maximum allowable fronthaul delay
$C_F^{(t)}$	required fronthaul capacity
$C_B^{(t)}$	required backhaul capacity
P_{ab}	path between node a and node b
Q	total delay

TABLE II: Latency and data-rate requirement of fronthaul for each slice.

Request	Type	Fronthaul Latency	Fronthaul Data-rate
URLLC	1	$\leq 50\mu$s	50-80 Mbps
eMBB	2	$\leq 100\mu$s	240-360 Mbps
mMTC	3	$\leq 250\mu$s	50-80 Mbps

IV. Results & Discussions

We consider a 100-node network with 50 CSs, 30 fog nodes, and 20 COs, with one core CO. The nodes are uniformly distributed over an area of 200 km^2. Each CO can serve a single BBU hotel. The nodes are connected by optical fiber cables, with a maximum of one fiber per link supporting eight 40 Gbps wavelengths. Furthermore, only traffic of the same type can be groomed into shared lightpaths, with four wavelengths dedicated to eMBB traffic and two each for URLLC and mMTC requests. We generate 3000 requests at random, with a predefined proportion of 1, 2, and 3 requests uniformly distributed across RRHs at CSs. Under split option 7, the required fronthaul capacity for type 1, type 2, and type 3 requests is distributed uniformly between $50 \sim 80$ Mbps, $240 \sim 360$ Mbps, and $50 \sim 80$ Mbps, respectively. Backhaul for requests of types 1 and 3 is uniformly distributed between $10 \sim 20$ Mbps, while requests of type 2 require 80 Mbps. Fronthaul delays for requests of types 1, 2, and 3 should not exceed 50μs, 100μs, and 250μs, respectively. We take into account propagation delays of 5μs/km, electronic switch delays of 20μs/switch, and interface delays of 25μs at RRH that occurs by split option 7. The fronthaul latency and data-rate requirements of all considered request types are summarized in Table II. Monte Carlo simulation is used to average the results of random simulations. The simulations are performed using Matlab on a system with an Intel(R) Core(TM) i7-9750H CPU @ 2.60 GHz with 16 GB RAM.

Figure 3 shows the number of BBU hotels and fog nodes with respect of changing proportion of all requests. The left plot in Fig. 3 shows the number of active nodes when ratio of URLLC requests is varied from 0 to 100, and the remaining requests are equally divided between eMBB and mMTC. As illustrated in the figure, no fog node is activated when there are no requests of type 1. In this case, only the BBU hotels are activated to serve type 2 and 3 requests. The number of active fog nodes increases as the proportion of type 1 requests rises. Furthermore, the number of active BBU hotels decreases as the proportion of type 2 and type 3 requests decreases in proportion to the increase in type 1 requests. Only fog

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.

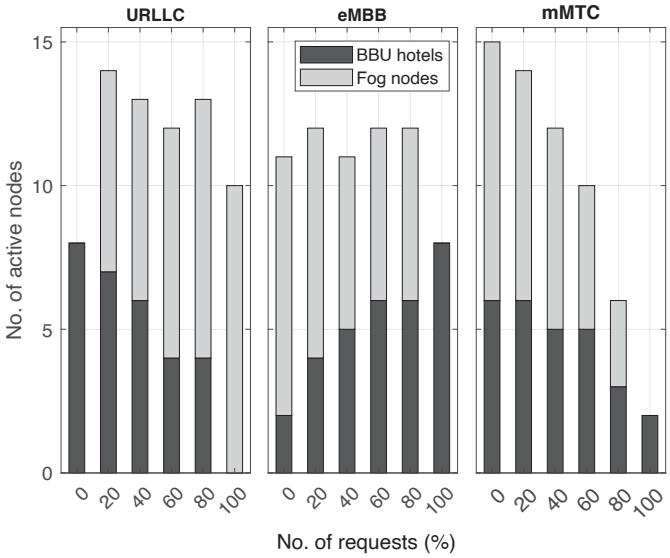


Fig. 3: Total number of active nodes w.r.t. total generated requests.

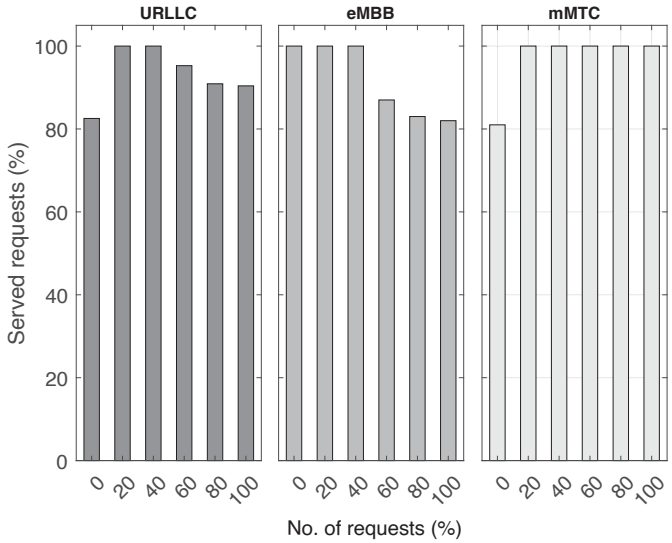


Fig. 4: Percentage of served request w.r.t. varying traffic.

nodes are activated when the proportion of requests of type 1 approaches 100%. The middle plot in Fig. 3 represents the case when the ratio of requests of type 2 varies with the remaining requests are distributed evenly between URLLC and mMTC. Since the eMBB requests are significantly more demanding in terms of bandwidth than mMTC requests, a greater number of BBU hotels are required to handle them as requests of type 2 increases. Moreover, as the proportion of type 1 traffic decreases, the number of fog nodes decreases. The right-most plot in Fig. 3 depicts the number of active nodes when type 3 requests are varied. In this instance, the number of BBU hotels and fog nodes simultaneously decreases as mMTC traffic increases. Since mMTC traffic is less bandwidth-intensive than eMBB traffic, an increase in mMTC requests results in the activation of a lower number of BBU hotels. In addition, the ratio of type 1 traffic is decreased as type 3 traffic increases, resulting in a decrease in the number of fog nodes.

Figure 4 depicts the percentage of served requests as the number of type 1, 2, and 3 requests varies. In the absence of type 1 requests, the percentage of served requests decreases, as the remaining 50% of requests are bandwidth-intensive eMBB requests. The percentage of requests served initially increases with an increase in type 1 traffic, then gradually declines. When the network is loaded with a large number of type 1 requests with stringent latency requirements, fog nodes will eventually be unable to cater 100% traffic. Furthermore, the percentage of served requests decreases as type 2 traffic increases. Since type 2 requests require significantly more capacity than type 1 and type 3 requests, an increase in the percentage of type 2 requests results in a lower percentage of served requests. Finally, as the ratio of type 3 requests increases, so does the number of served requests. The reason for this is that an increase in type 3 requests reduces type 1 and type 2 requests, which have stringent latency and bandwidth

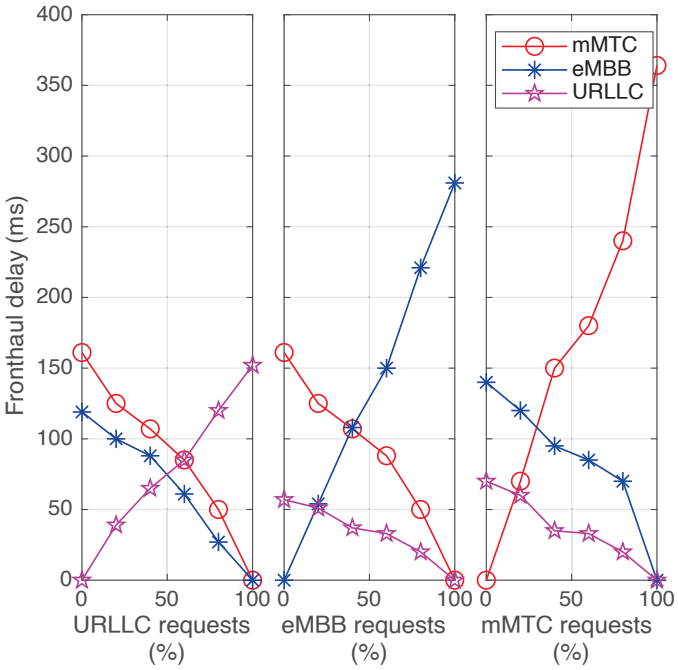


Fig. 5: Total fronthaul delay w.r.t varying traffic.

requirements than mMTC.

Figure 5 depicts the performance of each request type in terms of total fronthaul delay by varying the ratio of type 1, 2 and 3 requests in the incoming traffic. The total fronthaul delay for URLLC increases as type 1 traffic increases, while type 2 and 3 traffic decreases and eventually reaches zero. A similar trend is observed with an increase in the ratio of type 2 and 3 requests, which results in an increase in total fronthaul delay for the respective service, while the total fronthaul delay for the other two types decreases.

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this work, we implemented the network slicing paradigm in the a CF-RAN over WDM network architecture. We investigated the effect of variation in the ratio of slice requests due to the random nature of incoming traffic on the overall performance of the architecture. A heuristic algorithm is presented to evaluate the impact of traffic variation. The objective of the algorithm is to serve the maximum number of incoming requests by activating the minimum number of fog nodes/BBU hotels while considering the constraints for fronthaul latency and link capacity. Furthermore, the algorithm performs grooming, routing, and wavelength assignment (GRWA) for all three types of incoming requests.

Our results demonstrate that variations in the slice requests ratio of incoming traffic have a significant impact on network performance, and an architecture that can dynamically adapt to variations in traffic scenarios is a key enabler for network slicing in 5G. Specifically, we observed that an increase in the proportion of URLLC requests, with their stringent latency requirements, necessitates higher resource allocation at the fog nodes, potentially leading to resource contention and degraded performance for other services. The proposed heuristic algorithm addresses this challenge by dynamically adapting the resource allocation and routing strategies based on the observed slice request ratios. By minimizing resource utilization while maintaining maximum network connectivity, our algorithm ensures efficient and flexible resource management, enabling the CF-RAN architecture to accommodate diverse traffic scenarios effectively.

Acknowledgement This material is based upon work supported in part by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Early Career Research Program under Award Number DE-SC-0023957, and in part by the National Science Foundation under Grant No. 2323300.

References

[1] M. Alsenwi *et al.*, "eMBB-URLLC resource slicing: A risk-sensitive approach," *IEEE Commun. Lett.*, vol. 23, no. 4, pp. 740–743, 2019.

[2] N. Alliance, "5G white paper," *Next generation mobile networks, white paper*, vol. 1, no. 2015, 2015.

[3] J. Ordonez-Lucena *et al.*, "Network slicing for 5G with SDN/NFV: Concepts, architectures, and challenges," *IEEE Commun. Mag.*, vol. 55, no. 5, pp. 80–87, 2017.

[4] A. A. Barakabitze *et al.*, "5G network slicing using SDN and NFV: A survey of taxonomy, architectures and future challenges," *Comput. Netw.*, vol. 167, p. 106984, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1389128619304773

[5] L. M. Larsen *et al.*, "Fronthaul for cloud-RAN enabling network slicing in 5G mobile networks," *Wireless Commun. Mobile Comput.*, 2018.

[6] F. Musumeci *et al.*, "Optimal BBU placement for 5G C-RAN deployment over WDM aggregation networks," *J. Lightw. Technol*, vol. 34, no. 8, pp. 1963–1970, 2016.

[7] M. Ahsan *et al.*, "Functional split-aware optimal BBU placement for 5G cloud-RAN over WDM access/aggregation network," *IEEE Syst. J.*, pp. 1–12, 2022.

[8] M. Noweir *et al.*, "Linearization of radio-over-fiber Cloud-RAN transmitters using pre- and post-distortion techniques," *IEEE Photon. Technol. Lett.*, vol. 33, no. 7, pp. 339–342, 2021.

[9] A. Marotta *et al.*, "Reliable slicing with isolation in optical metro-aggregation networks," in *Opt. Fiber Commun. Conf. Exhib. (OFC)*, 2020, pp. 1–3.

[10] H. Yu *et al.*, "DU/CU placement for C-RAN over optical metro-aggregation networks," in *Intl. IFIP Conf. Opt. Netw. Design Model.*, 2019, pp. 82–93.

[11] S. Zhang, "An overview of network slicing for 5G," *IEEE Wireless Commun.*, vol. 26, no. 3, pp. 111–117, 2019.

[12] S. Costanzo *et al.*, "Dynamic network slicing for 5G IoT and eMBB services: A new design with prototype and implementation results," in *Cloudification Internet Things (CIoT)*, 2018, pp. 1–7.

[13] A. De Domenico *et al.*, "Optimal computational resource allocation and network slicing deployment in 5G hybrid C-RAN," in *IEEE Intl. Conf. Commun. (ICC)*, 2019, pp. 1–6.

[14] M. Masoudi *et al.*, "Energy-optimal end-to-end network slicing in cloud-based architecture," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 574–592, 2022.

[15] J. Tang *et al.*, "Fully exploiting cloud computing to achieve a green and flexible C-RAN," *IEEE Commun. Mag.*, vol. 55, no. 11, pp. 40–46, 2017.

[16] H. Yu *et al.*, "Isolation-aware 5G RAN slice mapping over WDM metro-aggregation networks," *J. Lightw. Technol.*, vol. 38, no. 6, pp. 1125–1137, 2020.

[17] C. Song *et al.*, "Hierarchical edge cloud enabling network slicing for 5G optical fronthaul," *J. Opt. Commun. Netw.*, vol. 11, no. 4, pp. B60–B70, 2019.

[18] S. Mondal and M. Ruffini, "Optical front/mid-haul with open access-edge server deployment framework for sliced O-RAN," *IEEE Trans. Netw. Service Manag.*, 2022.

[19] M. Ahsan *et al.*, "Efficient network slicing for 5G services in cloud fog-RAN deployment over WDM network," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 12 099–12 112, 2023.

[20] A. Ahmad *et al.*, "Traffic grooming and energy-efficiency in flexible-grid networks," in *IEEE Intl. Conf. Commun. (ICC)*, 2014, pp. 3264–3269.

[21] ——, "Switching node architectures in flexible-grid networks: A performance comparison," in *Intl. Conf. Opt. Netw. Des. Modeling (ONDM)*, 2014, pp. 49–54.

[22] L. M. Larsen *et al.*, "A survey of the functional splits proposed for 5G mobile crosshaul networks," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 146–172, 2018.

[23] F. Musumeci *et al.*, "Latency-aware CU placement/handover in dynamic WDM access-aggregation networks," *J. Opt. Commun. Netw.*, vol. 11, no. 4, pp. B71–B82, 2019.

Authorized licensed use limited to: UNIVERSIDADE DE SAO PAULO. Downloaded on October 05,2026 at 18:03:24 UTC from IEEE Xplore. Restrictions apply.